



ELSEVIER

Journal of Non-Crystalline Solids 291 (2001) 127–131

JOURNAL OF
NON-CRYSTALLINE SOLIDS

www.elsevier.com/locate/jnoncrsol

Ionic conductivity and relaxation in $\text{ZrF}_4\text{--BaF}_2\text{--YF}_3$ glasses: effect of substitution of BaF_2 by NaF

M. Sural, A. Ghosh *

Department of Solid State Physics, Indian Association for the Cultivation of Science, Jadavpur, Calcutta 700032, India

Received 5 April 2000; received in revised form 23 November 2000

Abstract

The ionic conductivity and relaxation of $50\text{ZrF}_4\text{--}(40-x)\text{BaF}_2\text{--}10\text{YF}_3\text{--}x\text{NaF}$ glasses for varying x have been investigated in the frequency range 10 Hz–2 MHz and in the temperature range from 300 K to just below the glass transition temperature. It is observed that the conductivity decreases with an increase in the NaF content, which is interpreted as arising from the trapping of F^- ions by Na^+ ions. The relaxation data have been analyzed in the framework of the electric modulus formalism. The values of the stretching exponent β obtained from the fits of the experimental data to the prediction of this formalism are almost independent of temperature and composition. The values of the decoupling index decrease with an increase of the NaF content, which is consistent with the decrease in the conductivity. © 2001 Elsevier Science B.V. All rights reserved.

1. Introduction

Fluoride glasses are of considerable interest because of their potential application in infrared optical components, ultra-low loss optical fibres, high power laser host materials, sensors, etc. [1,2]. Some fluoride glasses are likely candidate materials for solid electrolytes due to their relatively high electrical conductivity [3,4]. Ionic conduction in fluoride glasses formed with ZrF_4 as a glass former has been studied extensively [5–9]. The electrical conductivity of the alkali-free zirconium fluoride glasses arises mainly from the diffusion of fluorine ions [5]. However, fluoride glasses containing alkali ions as modifiers are fluorine and/or alkali ion conductors depending on the concentration and

the nature of alkali ions [6–11]. It has been observed that the electrical properties of the fluoride glasses containing lithium ions are different from that of the glasses containing sodium ions [10,11]. The ionic conductivity of the zirconium fluoride glasses containing lithium ions results from the diffusion of both F^- and Li^+ ions at low lithium concentration, while at higher lithium concentration the conductivity is due to the diffusion of Li^+ ions only. In contrast, the ionic conductivity of the zirconium fluoride glasses containing Na^+ ions arises solely from the diffusion of F^- ions for all concentrations of Na^+ ions [7]. These results have been confirmed by the ^{23}Na NMR and ^{23}Na -tracer diffusion studies [12,13]. In this paper, we have studied the ionic conductivity and relaxation of $50\text{ZrF}_4\text{--}(40-x)\text{BaF}_2\text{--}10\text{YF}_3\text{--}x\text{NaF}$ glasses for different NaF content (x). The objective is to observe the effect of the substitution of the primary modifier BaF_2 by NaF on the ionic relaxation.

* Corresponding author. Tel.: +91-33 473 4971; fax: +91-33 473 2805.

E-mail address: sspag@mahendra.iacs.res.in (A. Ghosh).

2. Experiment

Fluoride glass samples of compositions $50\text{ZrF}_4-(40-x)\text{BaF}_2-10\text{YF}_3-x\text{NaF}$ (Table 1) were prepared using the high purity chemicals ZrF_4 , BaF_2 , YF_3 and NaF . The appropriate proportions of these chemicals were mixed with an excess of ammonium bifluoride. The mixtures in a covered platinum crucible were melted at 800–900 °C for 15 min. The melts were then poured on to a preheated aluminum mould. Residual mechanical stresses were removed by annealing the samples at a temperature chosen 50 °C below the glass transition temperature T_g (Table 1) which was determined from the differential thermal analysis. Amorphous nature of the samples was confirmed from X-ray diffraction. For electrical measurements, gold electrodes were deposited on both surfaces of the polished samples of diameter ~ 10 mm and thickness ~ 1 mm by vacuum evaporation method. The gold coated samples were then heat treated at 150 °C for the stabilization of the electrode. Electrical measurements of capacitance and conductance of the samples were carried out in the frequency range 10 Hz–2 MHz using a QuadTech RLC meter (model 7600) interfaced with a computer. Measurements were made over a temperature range from 300 K to just below the glass transition temperature. The dc conductivity was obtained from the complex impedance plots.

3. Results

The dc conductivity obtained from the complex impedance plot is shown in Fig. 1 as a function of

reciprocal temperature for all glass compositions (Table 1). It is noted that the variation of the conductivity with temperature obeys an Arrhenius equation $\sigma = \sigma_0 \exp(-W_\sigma/kT)$. The values of the activation energy W_σ were obtained from the least squares straight line fits of the data and are shown in Table 1.

We have studied the ionic relaxation in the present fluoride glasses in the framework of the modulus formalism. In the modulus formalism [14], an electric modulus M^* is defined in terms of the reciprocal of the complex relative permittivity ϵ^* :

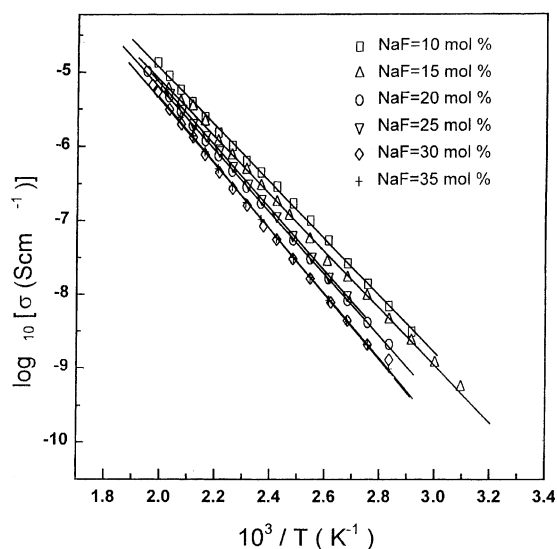


Fig. 1. The dc conductivity shown as a function of reciprocal temperature for the $\text{ZrF}_4\text{--BaF}_2\text{--YF}_3\text{--NaF}$ glasses for different compositions shown.

Table 1

Compositions, activation energies for the conductivity and the relaxation time, stretching exponent and high frequency dielectric constant for the $\text{ZrF}_4\text{--BaF}_2\text{--YF}_3\text{--NaF}$ glasses

Composition (mol %)				T_g (K) ± 5	W_σ (eV) ± 0.01	W_c (eV) ± 0.01	$\beta \pm 0.01$	ϵ_∞
ZrF ₄	BaF ₂	YF ₃	NaF					
50	30	10	10	584	0.74	0.74	0.44	12.4
50	25	10	15	579	0.76	0.77	0.47	12.5
50	20	10	20	583	0.83	0.83	0.48	14.2
50	15	10	25	531	0.83	0.84	0.48	13.4
50	10	10	30	547	0.86	0.88	0.48	13.3
50	5	10	35	533	0.87	0.87	0.46	11.9

$$M^* = 1/\varepsilon^* = M' + jM''$$

$$= M_\infty \left[1 - \int_0^\infty e^{-j\omega t} \{ -d\phi(t)/dt \} dt \right], \quad (1)$$

where M' and M'' are the real and imaginary parts of the complex modulus M^* , $M_\infty = 1/\varepsilon_\infty$ is the inverse of the high frequency dielectric constant ε_∞ and the function $\phi(t)$ gives the time evolution of the electric field within the material.

The frequency dependence of M' and M'' for a typical glass composition at several temperatures is shown in Figs. 2(a) and (b), respectively. It is noted that as the frequency is increased, M' shows dispersion and tends to saturate at M_∞ at higher frequencies. On the other hand, M'' shows an asymmetric maximum ($M''_{\max}$) centered at the dispersion region of M' . It is also observed in Fig. 2(b) that the maximum of M'' shifts towards higher frequencies as the temperature is increased. The angular frequency ω_c corresponding to $M''_{\max}$ gives the relaxation times τ_c from the condition $\omega_c \tau_c = 1$ [14]. We also observed a similar temperature and frequency dependence of M^* for other glass compositions.

4. Discussion

We observe in Fig. 1 that the conductivity decreases with an increase in NaF content. This behavior of these glasses is in sharp contrast with those of oxide glasses for which the conductivity increases with the increase in Na_2O content [15]. In the present glasses, in which BaF_2 is substituted by NaF, a fraction of the fluorine ions is trapped by the Na^+ ions. Consequently, an increase in the NaF content results in a decrease in the conductivity. This interpretation is consistent with the results of the ^{23}Na NMR and ^{23}Na -tracer diffusion studies [12,13].

A master plot of the modulus $M^*(\omega)$ at different temperatures is shown in Fig. 3 in which each frequency is scaled by ω_c and $M'(M'')$ by $M_\infty(M''_{\max})$ for the same glass composition as in Fig. 2. The perfect overlap of the curves for all temperatures into a single master curve indicates that the dynamical processes occurring at dif-

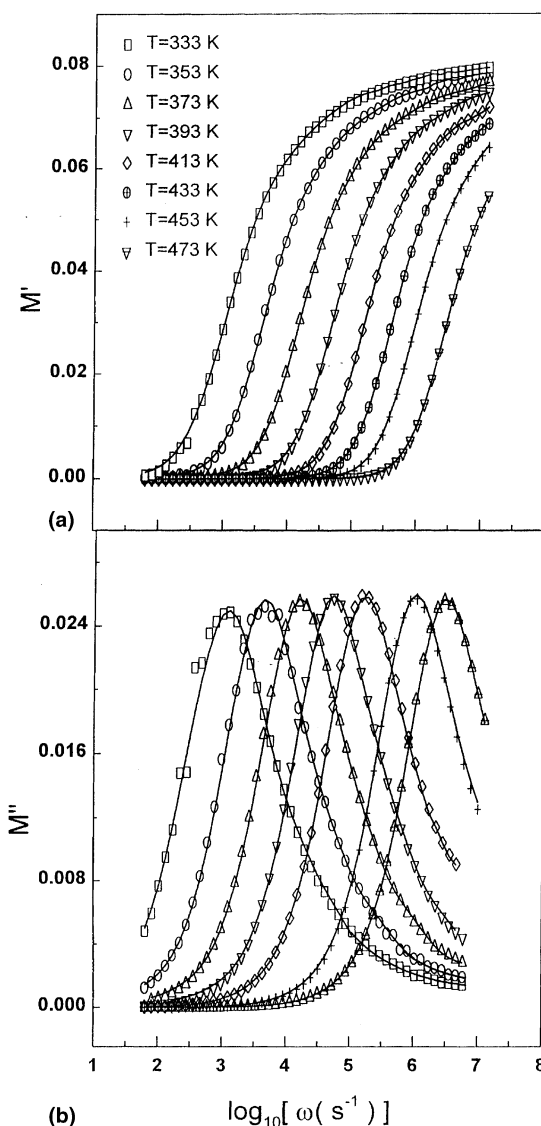


Fig. 2. (a) Frequency dependence of M' and (b) M'' at different temperatures (shown) for the $50\text{ZrF}_4\text{-}20\text{BaF}_2\text{-}10\text{YF}_3\text{-}20\text{NaF}$ glass composition. Solid curves are the best fits to the modulus formalism.

ferent frequencies are independent of temperature.

The data for M' and M'' for all temperatures were fitted simultaneously to the values obtained from Eq. (1). In the fitting process, we have used the Kohlrausch–Williams–Watts (KWW) function $\phi(t) = \exp[-(t/\tau_c)^\beta]$, where β is the stretched

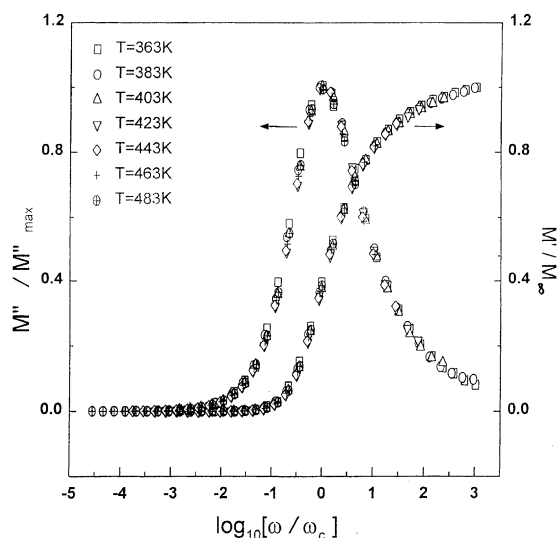


Fig. 3. Plots of M'/M_∞ and $M''/M''_{\max}$ versus $\log_{10}(\omega/\omega_c)$ at different temperatures for the same glass composition as in Fig. 2.

exponent indicating departure from the Debye relaxation. Best fits for M' and M'' at different temperatures for a glass composition are shown in Fig. 2. Other glass compositions also showed similar fits. The values of β and ϵ_∞ obtained from the fits are shown in Table 1, which indicates that the values of β for the present glass compositions are small and show a weak composition dependence. That the values of β are temperature independent is also evident from Fig. 3.

The temperature dependence of the conductivity relaxation time τ_c obtained from the maximum of M'' is shown in Fig. 4 for all glass compositions and it is evident that τ_c^{-1} shows an activated behavior and can be fitted to $\tau_c^{-1} = \tau_0^{-1} \exp(-W_c/kT)$. The values of the activation energy W_c obtained from the fits are shown in Table 1 for all glass compositions. It may be noted that the values of W_c are very close to the values of W_σ , which suggests that the ionic relaxation in the present glasses occurs by the hopping mechanism.

The composition dependence of the dc conductivity is consistent with that of the decoupling index. Angell [16] defined the glass decoupling index $R_\tau(T_g)$ as the ratio of $\langle\tau_s(T_g)\rangle/\langle\tau_c(T_g)\rangle$, where $\langle\tau_s(T_g)\rangle$ and $\langle\tau_c(T_g)\rangle$ are the average structural and

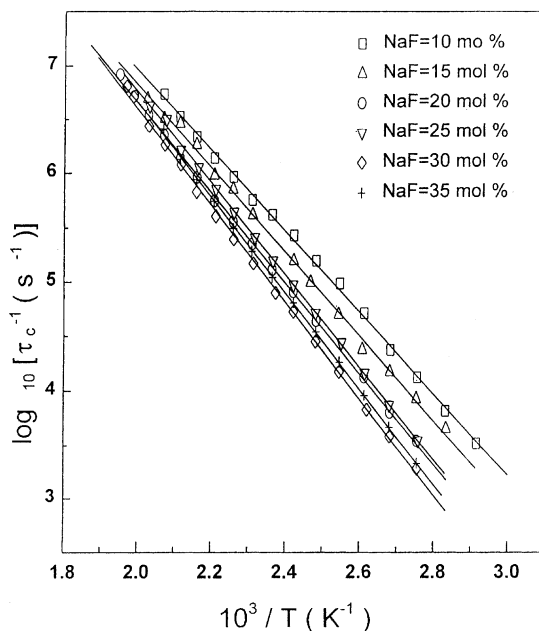


Fig. 4. The reciprocal of the conductivity relaxation time shown as a function of temperature for different composition of the $\text{ZrF}_4\text{-BaF}_2\text{-YF}_3\text{-NaF}$ glasses.

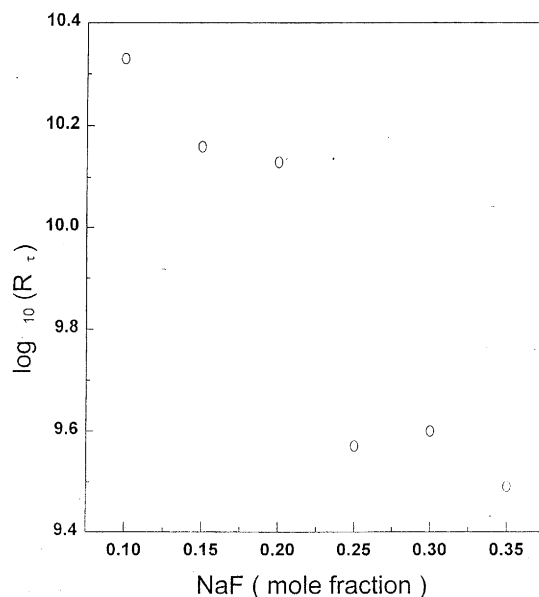


Fig. 5. Dependence of $R_\tau(T_g)$ on the NaF content in the composition for the $\text{ZrF}_4\text{-BaF}_2\text{-YF}_3\text{-NaF}$ glasses.

the conductivity relaxation times, respectively, at the glass transition temperature T_g . $R_\tau(T_g)$ determines the extent to which motion of the conducting ions is decoupled from the viscous motion of the glassy network and is related to the ability of the mobile ions to migrate in the glassy electrolytes at T_g . We have calculated $\tau_c(T_g)$ by extrapolating the plots of $\log_{10}(\tau_c^{-1})$ versus $10^3/T$ (Fig. 4) at T_g and assuming $\tau_s(T_g)$ equal to 200 s [16]. Fig. 5 shows the dependence of $R_\tau(T_g)$ on the NaF content. We observe that $R_\tau(T_g)$ decreases with the increase in NaF content, which suggests that the motion of the mobile F^- ions is decoupled less and less from the viscous motion of the glass network as the NaF content is increased in the composition suggesting a decrease in the conductivity with the increase of the NaF content.

5. Conclusions

The dc conductivity for ZrF_4 – BaF_2 – YF_3 –NaF glasses decreases, when BaF_2 is substituted by NaF, due to the trapping of F^- ions by the Na^+ ions. The result is consistent with the decrease in the decoupling index with an increase of the NaF content. The values of stretched exponent β obtained from the fit of the experimental data to the modulus formalism show weak composition dependence, but no temperature dependence. The comparable values of activation energies for dc conductivity and the relaxation time suggest that the ionic relaxation in the present glasses occurs by the hopping mechanism.

Acknowledgements

The financial assistance for this work by the Department of Science and Technology, Government of India (via the grant No. SP/S2/M26/93) is thankfully acknowledged.

References

- [1] J. Lucas, J. Mater. Sci. 24 (1984) 1.
- [2] M. Poulain, J. Non-Cryst. Solids 56 (1983) 1.
- [3] D. Ravaine, W.G. Perara, M. Miner, J. Phys. C: Solid State Phys. 9 (1982) 407.
- [4] C. Mai, M. Poulain, A.M. Asserio, G.P. Johari, J. Non-Cryst. Solids 113 (1989) 94.
- [5] J.M. Reau, J. Senegas, H. Aomi, P. Hagemuller, M. Poulain, Solid State Chem. 60 (1985) 159.
- [6] M. Sural, A. Ghosh, Solid State Ionics 120 (1998) 27.
- [7] M. Sural, A. Ghosh, J. Phys.: Condens. Matter 11 (1999) 3315.
- [8] J.M. Bobe, J.M. Reau, J. Senegas, M. Poulain, J. Non-Cryst. Solids 162 (1993) 169.
- [9] J.M. Bobe, J.M. Reau, J. Senegas, M. Poulain, J. Non-Cryst. Solids 209 (1997) 122.
- [10] J.M. Reau, J. Senegas, J.M. Rojo, M. Poulain, J. Non-Cryst. Solids 116 (1990) 175.
- [11] M. Sural, A. Ghosh, J. Phys.: Condens. Matter 10 (1998) 10577.
- [12] J.M. Reau, H. Khant, M. Poulain, J. Non-Cryst. Solids 11 (9) (1990) 347.
- [13] J.M. Bobe, J.M. Reau, J. Senegas, M. Poulain, Solid State Ionics 82 (1995) 39.
- [14] P.B. Macedo, C.T. Moynihan, R. Bose, Phys. Chem. Glasses 13 (1972) 171.
- [15] A. Pan, A. Ghosh, Phys. Rev B59 (1999) 899.
- [16] C.A. Angell, Chem. Rev. 90 (1990) 523.